12.4 Solving Problems With Lengths of Two Tangents

Lesson Introduction

Benchmark	MA.912.GR.6.1 Solve mathematical and real-world problems involving length of a secant, tangent, segment, or chord in a given circle.	Learning Target	• I can solve mathematical and real-world problems that involve lengths formed by the intersection of two tangents.

Benchmark Coherence	Building On	Working Toward
	N/A	N/A

Essential Questions	• How can you tell the difference between a secant and a tangent line? • What is the relationship between the tangent lines coming from the same point to a circle?
Lesson Narrative	Students will explore the relationship between intersecting tangent lines to a circle to discover that two tangent lines that intersect at the same external point are congruent. After discovering the relationship, they will apply that knowledge to solve for missing variables and segment lengths in mathematical and real-world problems.

Component Summary				
	12.4.1 Warm-Up (~5 min.)	Bell Work with secant and tangent lengths *(SE p. 247)*	12.4.3 Guided Practice (10-15 min.)	Solving problems involving intersecting tangent lines *(SE p. 248)*
	12.4.2 Exploration (10-15 min.)	Discovery of the tangent segment theorem *(SE p. 247)*	12.4.4 Your Turn! (10-15 min.)	Applying the tangent segment theorem *(SE p. 250)*
	12.4.5 Wrap Up (~5 min.)	Summarizing today's lesson in note to absent student *(SE p. 251)*		

Resources	Glossary	Materials	Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Glossary:</u> • Secant • Tangent • Congruent	• (Optional) Chart Paper • (Optional) Markers or Colored Pencils	• N/A

<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>Students may be confused between the tangent ratio from trigonometry and tangents of circles.</td><td>Emphasize the point of tangency and discuss the intersection of the radius to the point of tangency as perpendicular. As you illustrate this theorem, be sure to include a triangle whose sides are the radius of the circle, a line of tangency perpendicular to it, and a segment from the center to a point on the tangent line other than the outer endpoint of the radius.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Think-Pair-Share
12.4.2 Exploration is a collaboration opportunity for students. However, the questions are designed to spark thinking for the individual learner. Allow each student to take time to think about their answers, then share their responses with a partner, and finally with the whole class.</td><td>MA.K12.MTR.1.1: Participate in Effortful Learning
As students are working, continue to ask them questions that will allow them to analyze the problems in a way that makes sense to them. Encourage students to share thought processes with their partners and ask questions that will help with solving each problem.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">Consider combining the questions from 12.4.3 Guided Practice and 12.4.4 Your Turn!. Place students in small groups based on their levels of understanding. Each group will work through the questions using the instructional routine Take Turns. Each student will take a turn in completing a step of the question until the answer is found. This will encourage each student to participate in the components and provide students the opportunity to review work and ask clarifying questions, if needed.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
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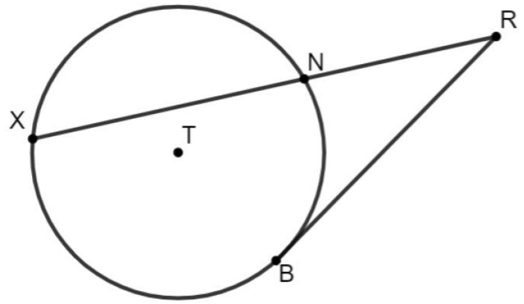
12.4.1 Warm-Up: Bell Work

Component Overview: *Students use prior knowledge of the intersection of secant and tangent segments to solve for a missing length.*

12.4.1 Warm-Up: Bell Work

1. Circle T has points X, N, and B on the circle. Secant $\overline{XR}$ and tangent $\overline{BR}$ intersect at point R outside the circle. Given that $XN = 16$ and $NR = 9$, find the length of $\overline{BR}$.



$(BR)^2 = 9(25), \quad BR = 15$

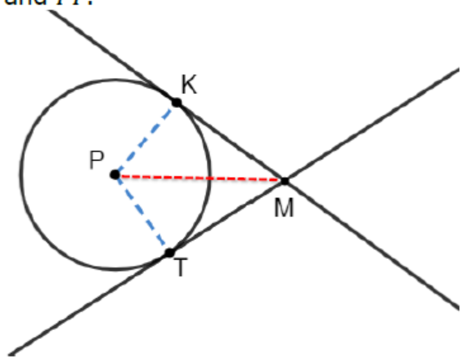
Consider using a Turn and Talk routine here to allow students to reflect and share their thoughts before completing this problem and then discussing whole-group. This allows for students who may struggle to articulate time to practice and verbalize their responses, increasing participation and confidence.

12.4.2 Exploration: Intersecting Tangent Lines

Component Overview: *Students explore the relationships between intersecting tangents lines by constructing a triangle to make a connection using prior knowledge of similar triangles. Students work with a partner to reveal that tangent lines to a circle from the same point are congruent.*

12.4.2 Exploration: Intersecting Tangent Lines

1. Circle P is shown with tangents $\overleftrightarrow{MK}$ and $\overleftrightarrow{MT}$ and radii $\overline{PK}$ and $\overline{PT}$.



   a. Draw a dotted line from point P to point M.
   See above.

   b. Discuss with your partner the relationships that can be determined given that $\overleftrightarrow{MK}$ and $\overleftrightarrow{MT}$ are tangents, and $\overline{PK}$ and $\overline{PT}$ are radii.
   $\angle MTP$ and $\angle MKP$ are right angles. Since $PK = PT$ and $PM = PM$, the triangles are congruent by the HL theorem.

   c. Discuss with your partner why HL congruence can be used to show that ΔMKP and ΔMTP are congruent.

   d. Since $\Delta MKP \cong \Delta MTP$, what is true about $\overline{MK}$ and $\overline{MT}$?
   They are congruent.

Tangent segments to a circle from the same point are congruent.

For more precise mathematical responses, consider a Notice and Wonder or Think-Pair-Share for questions 1b and 1c. Alternatively, post prompting questions to guide students to effective exploration:

- *What is formed by the line you drew in question 1a?*
- *What do you know about the relationship between a tangent line and the radius?*
- *What information do you know about the two triangles?*

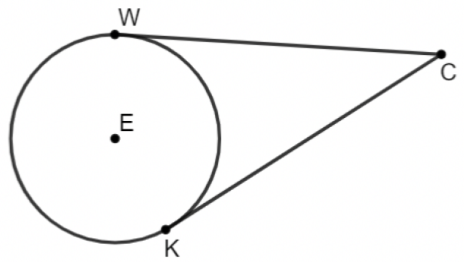
Listen for phrases like "point of tangency" or similar, and elicit any exemplar resources whole-group to synthesize their findings from 12.4.2 Exploration to apply in 12.4.3 Guided Practice.

12.4.3 Guided Practice: Solving Problems Involving Intersecting Tangent Lines

Component Overview: *Students begin to apply the intersecting tangent theorem in order to find the length of line segments and the value of variables.*

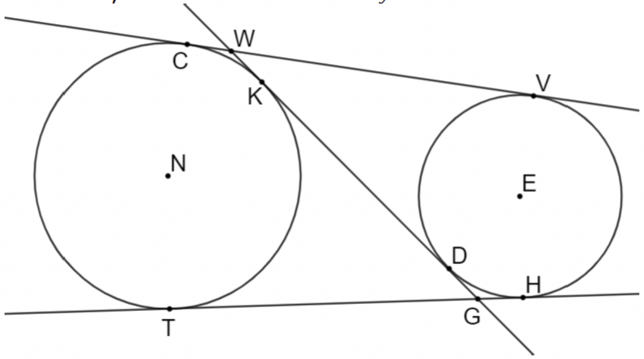
12.4.3 Guided Practice: Solving Problems Involving Intersecting Tangent Lines

1. Circle E has points W and K on the circle. Tangents $\overline{WC}$ and $\overline{KC}$ intersect at point C outside the circle. If $WC = x + 15$ and $KC = 2x - 5$, find the length of $\overline{KC}$.



$x + 15 = 2x - 5, \quad x = 20, \quad KC = 35$

2. Circle N has points C, K, and T on the circle. Circle E has points V, H, and D on the circle. Tangents $\overline{TG}$ and $\overline{KG}$ intersect outside the circle at point G. Tangents $\overline{VW}$ and $\overline{WD}$ intersect outside the circle at point W. If $TG = (7x - 36)$, $KG = (5x - 14)$, $VW = (8y + 13)$, and $WD = KG$, what is the value of y?



$5x - 14 = 7x - 36, \quad x = 11, \quad KG = 41$

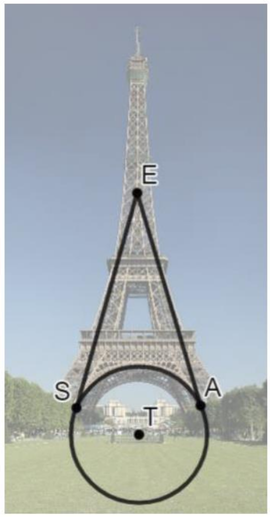
$41 = 8y + 13, \quad y = 3.5$

For students who may struggle, consider providing monomial expressions first before students solve a multi-step equation. Also, consider color coding, with colored markers or highlighters, each expression and tangent line for students to see.

What is the relationship between tangent line segments $\overline{TG}$ and $\overline{KG}$? What is the relationship between tangent line segments $\overline{KG}$ and $\overline{VW}$? How can you use these relationships to find the value of y?

12.4.3 Guided Practice: Solving Problems Involving Intersecting Tangent Lines (continued)

3. The Eiffel Tower in Paris, France stands at 324 meters tall. The tower has four semicircular arches at its base. Circle T represents one of the four arches at the base of the tower, where points S and A lie on the circle. If $SE = (17x + 27)$ meters (m) and $EA = (20x - 15)$ m, what is the length of $\overline{EA}$ in meters?



$17x + 27 = 20x - 15, \quad x = 14,$
$EA = 265\ m$

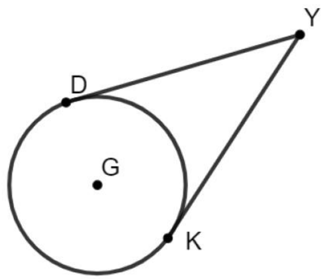
MA.K12.MTR.6.1: Assess the reasonableness of solutions and MA.K12.MTR.7.1: Apply mathematics to real-world contexts.
Provide opportunity for meaningful reflection with question 3. Applying mathematics to real-work contexts will help connect mathematical concepts to everyday experiences. This question will also challenge students to question the accuracy of the model and the method they used.

12.4.4 Your Turn!

Component Overview: *Students are finding the value of variables by solving questions involving intersecting tangent lines. Students may engage with this component independently or peer review their partner's responses for additional error analysis practice.*

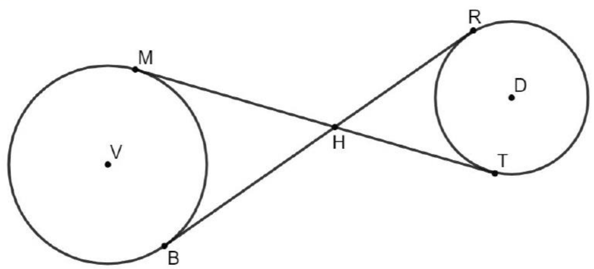
12.4.4 Your Turn!

1. Circle G has points D and K on the circle. Tangents $\overline{DY}$ and $\overline{KY}$ intersect at point Y outside the circle. If $DY = 4x - 19$ and $KY = 15$, find the value of x.



$4x - 19 = 15, \quad x = 8.5$

2. Points M and B are on the circle V and points R and T are on the circle D. $\overline{MT}$ and $\overline{BR}$ intersect at point H. $\overline{MT}$ is tangent to circle V at point M and tangent to circle D at point T. $\overline{BR}$ is tangent to circle V at point B and tangent to circle D at point R. If $MH = (4x - 8)$, $BH = (2x + 1)$, $TH = (2x - 1)$, and $RH = (5y - 7)$, what is the value of y?



$2x + 1 = 4x - 8, \quad x = 4.5$

$2(4.5) - 1 = 5y - 7, \quad y = 3$

Clarify, Critique, Correct
Have students exchange their work to analyze the work of their partner. This routine fortifies output and engages students in meta-awareness.

12.4.5 Wrap-Up: Cool Down

Component Overview: *Students take time to reflect by writing a short note summarizing the lesson.*

12.4.5 Wrap-Up: Cool Down

1. Write a short note summarizing today's lesson to a friend who missed class.

Answers will vary.

Discuss any questions the students still have regarding the content. Consider a Parking Lot approach with a list of the questions on the board or poster in which students can either answer each other's questions or you can revisit as a class later.